

Wearable solutions

Wearable technology is one of the most reliable and cost-effective safety solutions, especially for safekeeping on-field workforce and resources. V.S. Shridhar, senior vice president and head - Internet of Things, Tata Communications, says, "Wearable sensors provide a non-intrusive, hands-free, always-on, environment-aware solution that offers direct delivery of critical information to those who need it and at the time they need it."

He adds, "These help address concerns like employee visibility and traceability in emergency situations, alerts during accidents, movements in restricted areas, stress levels, women's safety during commute and more. These also enable safety supervisors in such environments, with timely updates, to improve the response time to incidents and predict future behaviour."

Tata Communications was one of the first Indian organisations to design and deploy safety technology. It designed smartwatches for industrial workers that can measure real-time health parameters and surrounding environmental conditions.

Shridhar says, "Tata Communications' smart wearable for Employee Health & Safety (EHS) help enterprises monitor employee wellness by tracking their vital stats, generate alerts on detection of anomalies, falls, geo-fenced location triggers or any form of SOS communication from the employees through a button press. Through a managed service and cloud-based application that delivers actionable insights, the wearable enables decision makers to transition from preventive to predictive safety."

Stored data can be analysed to produce strategies against risky operation processes. Additionally, it can regulate and restrict unauthorised access. Shridhar adds, "Maximum impact of such solutions is seen in industries where a large number of the workforce is deployed in potentially hazardous environments including manufacturing plants, construction, mining, healthcare, disaster response and recovery, and off-shore drilling."

NXP India recently provided their female work strength with a smart



Universal Robots UR10e robot arm and base
(Credit: www.universal-robots.com)

tracker called SAFER. The pendant-like device acts as an emergency beacon that can share urgent notifications with real-time location details to stored contacts. Recipient smartphones can track the location of the wearer. Non-smartphone cellular devices receive text-based updates on the loca-

tion. The tracker also helps connect to nearby medical facilities and police stations. This initiative was taken to enhance women's safety, especially when they travel late after work.

Collaborative machines

In industrial applications, collaborative robots (in short, cobots) are slowly replacing traditional robots in compatible operations. Pradip David, general manager - South Asia, Universal Robots, says, "Cobots are basically force-limiting robots. Powered by precise sensors, these smart machines can sense a nearby person or an external force, and stop immediately without impact. These can be set between 50 Newton and 250 Newton force threshold."

"When force exerted on a person exceeds the threshold, the machines go into a protective stop. Operation can be restored from the same spot later. Thus, cobots do not require protective fences."

"Safety comes at a trade-off of speed and payload. Payload is restricted to 10kg."

The main motive of cobots is to decrease accidents in industries and increase human-machine collaboration. These can replace traditional robots for many automated operations, like pick-and-place, packaging, palletising, injection moulding and so on. Simultaneously, these can reduce running costs and improve factory productivity.

David explains, "Traditional robots have regular maintenance expenses, including oil, grease and battery changes. All this amounts up to ₹ 100,000 annually, at least. The manufacturing line may need to be shut down for the time these robots are serviced, creating a lengthy downtime. Moreover, these draw huge amounts of energy as compared to cobots."

"Cobots, on the other hand, are fix-and-forget devices. For a cobot, a typical ROI can take up to two to three years, depending on the scale of requirement."

As an example, SMEW Textile Machinery Pvt Ltd, an Ahmedabad-based textile machinery manufacturing enterprise, installed a cobot from Universal Robots for pick-and-place operations in January 2017. The enterprise reports that its production improved three times (from 30 pieces output to 80-90 pieces per week), risks of accident reduced significantly and it got an ROI within one year.

Collateral benefits

Smart safety solutions can reduce 40 to 60 per cent expenses on mishap compensations. Shridhar mentions, "These solutions also aid in boosting employee morale and motivation, reduce absenteeism and increase productivity of the workforce. These automate headcount and improve regulatory compliance. The benefits cumulatively lead to reduction in operating costs and improved employee satisfaction index."

India tracks high labour accidents in workplaces. Considering all the benefits and solutions available out there, it is a good time for employers to take a closer look at the security of their workforce.

—Paromik Chakraborty, technical journalist, EFY

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